

Pandas Data Analysis Assessment

Objective:

Test your skills in **data exploration, manipulation, aggregation, and visualization** using Pandas, NumPy, and Matplotlib.

Dataset:

Download [Dataset](#)

Columns: gender, race/ethnicity, parental level of education, lunch, test preparation course, math score, reading score, writing score

Instructions:

- Import the necessary libraries (pandas, numpy, matplotlib.pyplot)
 - Submit a **Jupyter notebook** with all code, outputs, and answers.
 - Comment your code to explain your steps.
 - Include titles and axis labels in all plots.
 - [Submission link](#)
-

Section A – Basic Exploration

1. Read the StudentsPerformance.csv dataset.
 2. Display the **first 10 rows and the last 10 rows**.
 3. Check the dataset overview using `.info()`.
 4. Check the shape of the dataset.
 5. List all column names.
 6. Identify which columns are **numerical** and which are **categorical**.
 7. Display descriptive statistics for numerical columns using `.describe()`.
 8. Check if there are any **missing values** in the dataset.
-

Section B – Filtering & Conditional Selection

1. Select all students who completed the test preparation course.
2. Find all students who scored above 70 in math.
3. Retrieve female students who scored at least 65 in all three subjects (use the `&` operator).
4. Count how many students have free/reduced lunch.

Section C – GroupBy & Aggregation

1. Compute average math score by gender.
 2. Compute average scores (math, reading, writing) by parental level of education.
 3. Compare average scores of students with vs without test preparation (note: 'none' refers to students who did not complete the test preparation course).
-

Section D – Visualization

Instructions: Use Matplotlib. Include titles and axis labels.

1. Plot histograms for:
 - math score
 - reading score
 - writing score
 2. Create a bar plot showing average score by gender.
-

Section E – Interpretation

1. In 3–5 sentences, answer:
 - Does test preparation appear to improve performance?

Notes for Students

- Ensure your code runs without errors.
- Use comments to explain your steps.
- Label all plots properly.